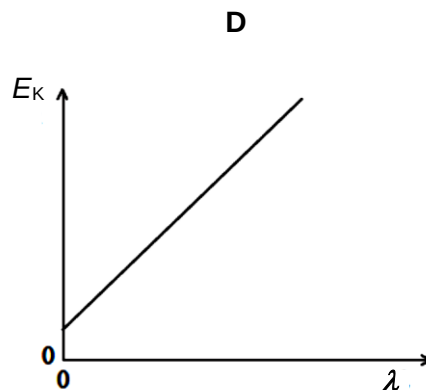
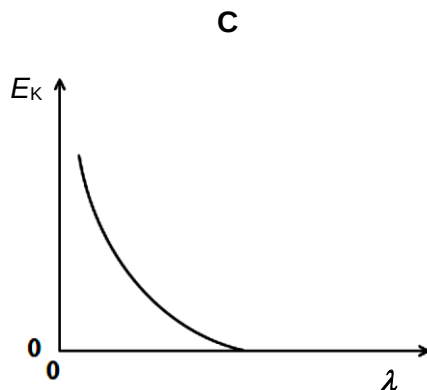
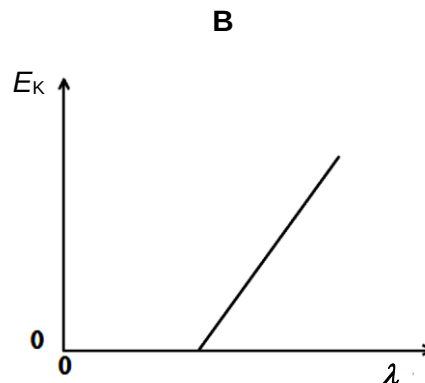
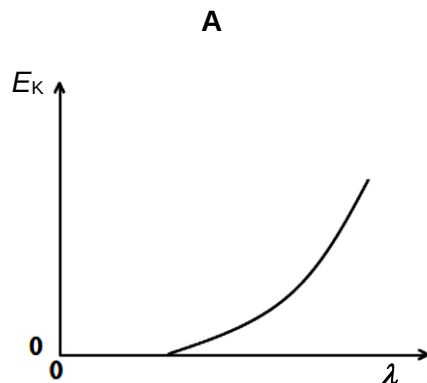


- 28 Light of wavelength λ is incident on a metal surface in a vacuum. Photoelectrons are emitted from the surface of the metal.

Which of the following best shows the variation with λ of the maximum kinetic energy E_K of the emitted electrons?



Ans: C

$$\frac{hc}{\lambda} = \phi + E_K$$

$$E_K = \frac{hc}{\lambda} - \phi$$

Plotting E_K against λ gives a graph of $y = 1/x$ (both axes as asymptotes) which is then shifted downwards to touch the horizontal axis.

- 29 An atom makes a transition from an energy level of energy E_1 to a level of energy E_2 emitting